4. **Soil Temperature**-Soil gets its heat energy from solar radiation, decomposing organic matter, and heat from the interior of earth. Soil temperature effects the germination of seeds, growth of roots and biological activity of soil-inhabiting micro-and macro-organisms.
5. **Soil water**- In soil, water is not only important as a solvent and transporting agent, but also maintains soil texture, arrangement and compactness of soil particles, making soil habitable for plants and animals.

### Wind

Wind is the natural movement of air of any velocity from a particular direction. The two main causes are differential heating between the equator and the poles and the rotation of the planet (Coriolis effect). Wind helps to transport pollen grains, seeds, and even flight of birds. While it is the source of wind energy, it also causes erosion. Wind speed is measured with an Anemometer.

### Humidity

Moisture in the form of invisible vapor in the atmosphere is called humidity. which is generally expressed in terms of absolute humidity, relative humidity or specific humidity. Absolute humidity is the total mass of water vapour present in a given volume or mass of air. It does not take temperature into consideration.

Relative humidity is the amount of water vapour present in air and is expressed as a percentage of the amount needed for saturation at the same temperature Relative humidity is expressed as a percentage; a high percentage means that the air-water mixture is more humid at a given temperature. Humidity is measured with a Hygrometer.

### Altitude

This factor is mainly the elevation or gradient and it affects temperature and precipitation in an ecosystem or biome. As altitude increases, temperature and density of oxygen decreases. Higher altitudes usually receive snow instead of rain because of low temperature.

Animals are known to modify their response to environmental changes (stress) in reasonably short time spans. This is known as **Acclimatization**. This is observed when people who have moved from the plains to higher altitudes show enhanced RBC count within a few days of settling in their new habitat. This helps them cope with lower atmospheric oxygen and higher oxygen demand.

## 11.4 Concept of biome and their distribution

Biomes are large regions of earth that have similar or common vegetation and climatic conditions. They play a crucial role in sustaining life on Earth. They are defined by their soil, climate, flora and fauna. Biomes have distinct biological communities that have been formed in response to a shared physio-chemical climate. Biomes are seen to even spread across continents. Thus, it can be observed that a biome is a broader term than habitat. Any biome can comprise a variety of habitats. Factors such as temperature, light, water availability determine what type of organisms and adaptations are observed in a biome (**Fig. 11.4).**

### Characters of a biome

- Location, Geographical position (Latitude, Longitude)
- Climate and physiochemical environment
- Predominant plant and animal life
- Boundaries between biomes are not always sharply defined. Transition or transient zones are seen as in case of grassland and forest biomes. (**Fig. 11.5)**

### Aquatic Biomes

They occupy about 71% of the biosphere. The aquatic biome is home to millions of aquatic organisms like fishes. The climate of coastal zones are influenced by aquatic bodies.



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